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Chapter 1

We gotta start somewhere

Given a finite connected graph K and it vertices V_m the for a function $u : V_m \rightarrow \mathbb{R}$ we define the energy to be

$$E_K(u) = \sum_{x \sim y} (u(x) - u(y))^2$$

Theorem 1.1: Let $u \in \text{dom } \Delta_\mu$. Then the pointwise formula

$$\Delta_\mu u(x) = \lim_{m \rightarrow \infty} r^m \left(\int_K \psi_x^{(m)} d\mu \right)^{-1} \delta_m u(x)$$

holds with the limit uniform across $V_* \setminus V_0$. Conversely, suppose u is a continuos function and the right side of the limit converges uniformly to a continuos function on $V_* \setminus V_0$. Then $u \in \text{dom } \Delta_\mu$ and the equation holds.

Proof. Suppose $u \in \text{dom } \Delta_\mu$ and $\Delta_\mu u = f$, then by [\[INSERT REF\]](#) we have that

$$r^{-m} \left(\int_K \psi_x^{(m)} d\mu \right)^{-1} \Delta_m u(x) = \frac{\int_K f \psi_x^{(m)} d\mu}{\int_K \psi_x^{(m)} d\mu}$$

for any $x \in V_m \setminus V_0$. Uniform converges of the right side is because of continuity of $f(x)$. Since

$$\begin{aligned} \lim_{m \rightarrow \infty} \int_K f \psi_x^{(m)} d\mu &= \int_K f \lim_{m \rightarrow \infty} \psi_x^{(m)} d\mu \\ &= \int_K f \delta_{xy} d\mu \\ &= f(x) \int_K \delta_{xy} d\mu \end{aligned}$$

in which case

$$\lim_{m \rightarrow \infty} r^{-m} \left(\int_K \psi_x^{(m)} d\mu \right)^{-1} \Delta_m u(x) = \lim_{m \rightarrow \infty} \frac{\int_K f \psi_x^{(m)} d\mu}{\int_K \psi_x^{(m)} d\mu} = f(x) = \Delta_\mu u(x)$$

Moving on to the second half of the proof. Suppose that $r^m \left(\int_K \psi_x^{(m)} d\mu \right)^{-1} \delta_m u(x)$ converges uniformly to a continuos function f . Now for any $v \in \text{dom}_0 \mathcal{E}$ we have

$$\mathcal{E}_m(u, v) = r^{-m} \sum_{\substack{x \sim y \\ m}} (u(y) - u(x)) (v(y) - v(x)).$$

If we collect all the terms that contain the factor $v(x)$ for a fixed $x \in V_m \setminus V_0$ then each $v(x)$ will be of the form $-v(x) \sum_m (u(y) - u(x))$ for each $y \sim x$. In which case we get

$$\begin{aligned} \mathcal{E}_m(u, v) &= -r^{-m} \sum_{x \in V_m \setminus V_0} v(x) \left(\sum_{\substack{y \sim x \\ m}} (u(y) - u(x)) \right) \\ &= - \sum_{x \in V_m \setminus V_0} v(x) r^{-m} \Delta_m u(x) \end{aligned}$$

While it seems we lose many terms by removing $v(y)$, but they are instead contained in $\sum_{\substack{y \sim x \\ m}} (u(y) - u(x))$. Note that the sum is "reset" for every x so we count both the relation $x \sim y$ and $y \sim x$.

Now if we define $f_m(x) = r^{-m} \left(\int_K \psi_x^{(m)} d\mu \right)^{-1} \Delta_m u(x)$ then $f_m \rightarrow f$ uniformly by assumption and we can rewrite the above as

$$\begin{aligned} \mathcal{E}_m(u, v) &= - \sum_{x \in V_m \setminus V_0} v(x) \left(\int_K \psi_x^{(m)} d\mu \right) r^{-m} \left(\int_K \psi_x^{(m)} d\mu \right)^{-1} \Delta_m u(x) \\ &= - \int_K \left(\sum_{x \in V_m \setminus V_0} v(x) f_m \psi_x^{(m)} \right) d\mu. \end{aligned}$$

Looking at the sum in the above, we find the following

$$\lim_{m \rightarrow \infty} \sum_{x \in V_m \setminus V_0} v(x) f_m(x) \psi_x^{(m)} = \lim_{m \rightarrow \infty} \sum_{x \in V_m \setminus V_0} v(x) f_m(x) \psi_x^{(m)}$$

□

Bibliography

- [Str06] Robert S. Strichartz. *Differential Equations on Fractals: A Tutorial*. Princeton University Press, 2006.